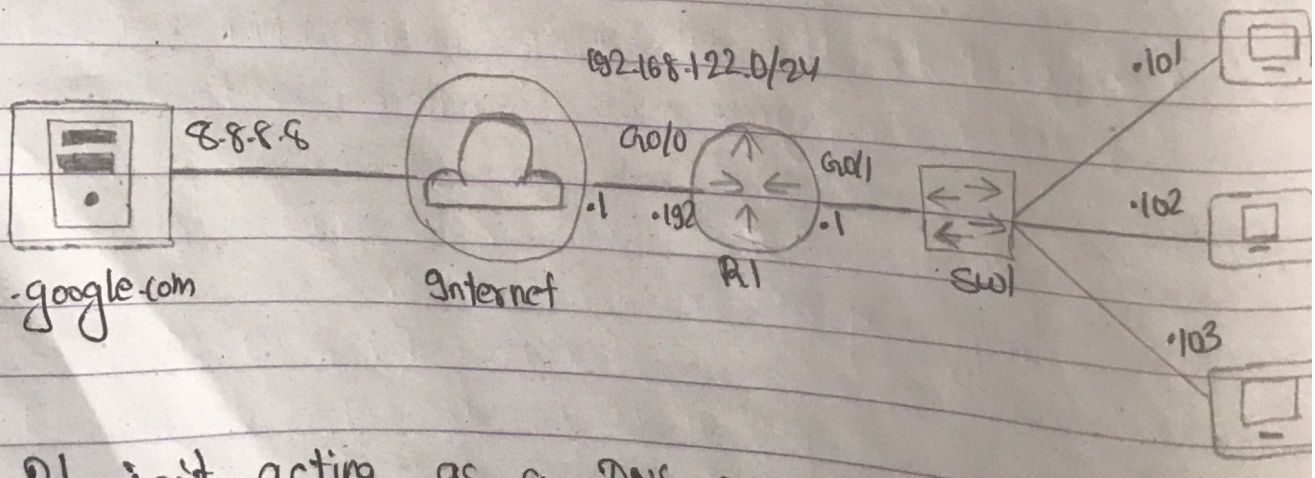
Day-38

Domain Name System (DNS)

Purpose of DNS.

- DNS is used to resolve human-readable names (google.com) to IP addresses.
- Machines use such as PCs don't use names, they use addresses (i.e IPv4/IPv6).
- Names are much easier for us to use and remember than IP addresses.
  ↳ What is the IP addresses of youtube.com?
- When we type 'youtube.com' into a web browser, ~~by~~ our device will ask a DNS server for the IP address of youtube.com
- The DNS server(s) our device uses can be manually configured or learned via DHCP.

Network Topology.



↳ R1 isn't acting as a DNS server or client. It is simply forwarding packets. No DNS configuration is required on R1.

Feco Windows Command

i) ipconfig /all

DNS Servers . . . . . : 8.8.8.8

ii) nslookup youtube.com.

It tells the device to ask its DNS server for IP address of the specified name

Non-authoritative answer: Name: youtube.com

Addresses: IPv6 add /n IPv4 address

* DNS 'A' record = Used to map names to IPv4 addresses
* DNS 'AAAA' record = Used to map names to IPv6 addresses
* DNS uses Both UDP/TCP.
* Standard DNS queries/responses typically use UDP. TCP is ~~used~~ used for DNS messages greater than 512 bytes. In either case port 53 is used.

DNS Cache

* Devices will save the DNS server's responses to a local DNS cache. This means they don't have ~~the~~ to query the server every single time they want to access a particular destination.

To view dns cache on windows PC.
Command: ipconfig /displaydns

To clear dns cache on windows PC.
command: ipconfig /flushdns.

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On Hosts file

On addition to DNS cache, most devices have hosts file.
On linux at /etc/hosts & in windows at c:\Windows\system32\drivers\etc\hosts

Eg:
It's a list of hosts & IP addresses.
192.168.1.1 R1
now when we ping R1, 192.168.1.1 will be pinged

DNS in Cisco IOS

- For hosts in a network to use DNS, we don't need to configure DNS on the routers. They'll simply forward the DNS messages like any other packets.

- However, a Cisco IOS can be configured as a DNS server, although its rare.
  ↳ If an internal DNS server is used, its usually a Windows or Linux server.

- A CISCO router can be configured as DNS client, so we can execute PING and other commands using names instead of IP add.

Router as DNS Server

```
R1(config) # ip dns server
R1(config)# ip host R1 192.168.0.1
R1 (config)# ip host PC1 192.168.0.101
R1 (config)# ip host PC2 192.168.0.102
R1 (config)# ip host PC3 192.168.0.103
```

[Configure R1 to act as DNS server]

host table

R1(config)# ip name-server 8.8.8.8 [Configure a DNS server that R1 will query if the requested record isn't in its host table]

a)* R1(config)# ip domain lookup

[Enable R1 to perform DNS queries. (enabled by default) Old version of the command is 'ip domain-lookup']

If (a)* command isn't enabled, R1 won't be able to query Google's DNS Server.

To view both hosts configured as well as hosts learned from and cached via DNS, use

R1# Show hosts

CISCO router as DNS client.

R1(config)# ip name-server 8.8.8.8

R1(config)# ip domain-lookup

Optional command:

R1(config)# ip domain name jeremysitlab.com

- Configures the default domain name
- This'll be automatically appended to any hostnames without a specified domain.

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